

HOSPITAL OPERATIONS & PATIENT ANALYTICS REPORT

Problem Statement:

Hospitals generate large amounts of patient and operational data. Raw data can be difficult to analyse when it contains duplicate records, missing values, and inconsistent entries.

This project focuses on cleaning and analysing hospital data to understand patient volume, departmental activity, billing, demographics, waiting time, satisfaction, and readmission patterns. The analysis uses Excel, SQL, Python, and Power BI to turn the raw data into useful insights.

1. Project Objectives

The main objectives of this project are:

1. Clean and prepare the raw hospital dataset by handling duplicate records, missing values, and data inconsistencies.
2. Analyse patient volume across hospital departments and identify departments with higher patient activity.
3. Examine monthly patient admission trends over the 2024–2025 period.
4. Analyse hospital billing patterns and compare average treatment costs across departments.
5. Understand patient demographics, including age groups and gender distribution.
6. Examine the relationship between waiting time and patient satisfaction.
7. Analyse readmission patterns and discharge outcomes.
8. Perform exploratory data analysis using Python to identify statistical patterns and relationships.
9. Develop an interactive Power BI dashboard to present important hospital operational and patient-related metrics.
10. Generate data-driven findings and recommendations that can support better understanding of hospital operations and patient experience.

2. Business Questions

The analysis was designed around the following business questions:

1. How many patient records are available after data cleaning?
2. Which hospital departments have the highest patient volume?
3. How do patient admissions change month by month?
4. What is the average hospital bill across all patient records?
5. Which departments have the highest average hospital treatment costs?
6. What are the most common admission types?
7. Which medical conditions are most frequently recorded?
8. How are hospital costs distributed between insurance coverage and patient-paid amounts?
9. What proportion of patients are readmitted?
10. What are the different discharge outcomes recorded in the hospital dataset?

11. How are patients distributed across different age groups?
12. Is there a relationship between patient waiting time and satisfaction?
13. Is there a relationship between length of stay and total hospital billing?

These questions were used to guide the SQL analysis, Python exploratory analysis, and Power BI dashboard development.

3. Dataset Overview

The dataset contains hospital patient and operational records for the period from 2024 to 2025.

- Original Records: 30,075
- Final Cleaned Records: 30,000
- Columns: 24

The dataset includes information about patients, departments, medical conditions, admission details, waiting time, billing, insurance, satisfaction, readmission, and discharge status.

4. Tools Used

- Excel – Data cleaning and preparation
- SQL (PostgreSQL) – Data analysis
- Python – Exploratory Data Analysis
- Power BI – Dashboard and visualization

5. Excel Data Cleaning

The raw dataset was first cleaned and prepared in Excel.

The following steps were performed:

- Removed duplicate records using Record_ID.
- Handled missing values in important columns.
- Replaced missing Payment Mode and City values with **Unknown**.
- Filled missing Satisfaction Rating values with the median value **4.6**.
- Checked text fields for consistency.
- Verified date and numeric column formats.
- After cleaning, the final dataset contained **30,000 records**.

The cleaned dataset was then used for SQL, Python, and Power BI analysis.

6. SQL Analysis

SQL was used to answer key business questions from the cleaned hospital dataset.

- Total patient records?
30,000
- Which department has the highest patient volume?
General Medicine – 5,364 records
- What is the average hospital bill?
₹45,226.92
- Which department has the highest average bill?
Cardiology – ₹73,942.90
- What is the most common admission type?
Routine – 18,573 records
- What is the most common medical condition?
Fever – 2,678 records
- Total insurance coverage?
₹425,476,925.56
- Total patient-paid amount?
₹929,708,770.72
- What is the readmission rate?
15.14%
- What is the most common discharge status?
Recovered – 13,921 records

7. Python Exploratory Data Analysis

Python was used to explore the cleaned dataset and identify important patterns.

Key Findings

- Patients aged **66+** formed the largest age group with **11,424 records (38.08%)**.
- Monthly admissions remained relatively stable during 2024–2025.
- **July 2025** recorded the highest monthly admissions with **1,354 records**.
- **February 2025** recorded the lowest monthly admissions with **1,098 records**.
- The average patient satisfaction rating was **4.49 out of 5**.
- Waiting time and satisfaction showed a **moderate negative correlation (r = -0.39)**.
- Length of stay and total hospital bill showed a **moderate positive correlation (r = 0.55)**.

- Statistical outliers were identified using the IQR method, but they were **not removed**, as they may represent genuine hospital cases.

8. Power BI Dashboard

An interactive dashboard was created in Power BI to present the main hospital performance and patient-related metrics.

Dashboard Includes

- Total Patient Records
- Average Hospital Bill
- Average Patient Satisfaction
- Readmission Rate
- Monthly Patient Admissions
- Patient Records by Department
- Average Hospital Bill by Department
- Patient Records by Age Group
- Average Satisfaction by Waiting Time
- Admission Type Distribution

The dashboard helps to view important hospital trends and metrics in an easy-to-understand visual format.

9. Key Findings & Insights

- The dataset contains **30,000 cleaned patient records**.
- **General Medicine** recorded the highest patient volume with **5,364 records**.
- **Cardiology** had the highest average hospital bill at **₹73,942.90**.
- The average hospital bill across all records was **₹45,226.92**.
- Patients aged **66+** formed the largest age group with **38.08%** of records.
- The average patient satisfaction rating was **4.49 out of 5**.

- Waiting time showed a **negative correlation with satisfaction** ($r = -0.39$).
- Length of stay showed a **positive correlation with total bill** ($r = 0.55$).
- The overall readmission rate was **15.14%**.

10. Recommendations

Based on the analysis, the following actions can be considered:

- **Reduce waiting time:** Since waiting time has a negative relationship with patient satisfaction, hospitals can focus on reducing unnecessary waiting, especially during busy periods.
- **Plan resources by department:** General Medicine has the highest patient volume, so patient flow and resource requirements can be planned accordingly.
- **Review high-cost departments:** Cardiology and Neurology have higher average bills. Their treatment costs can be reviewed to better understand the major cost drivers.
- **Use admission trends for planning:** Monthly admission patterns can be used to support staff and resource planning.
- **Monitor readmissions:** The 15.14% readmission rate can be tracked regularly to identify patterns and understand possible areas for further investigation.
- **Focus on older patients:** Since patients aged 66+ form the largest age group, hospital services and resource planning can consider the needs of older patients.

11. Conclusion

This project analysed hospital patient and operational data using Excel, SQL, Python, and Power BI.

The analysis provided insights into patient volume, department performance, hospital billing, patient age groups, waiting time, satisfaction, and readmission patterns.

The Power BI dashboard brings the main findings together in an interactive and easy-to-understand format. Overall, the project demonstrates how raw hospital data can be cleaned, analysed, and transformed into useful insights for understanding hospital operations and patient experience.